



THE AI TOOLBOX · CLASS 15

Research Showcase, Defense & Portfolio Launch

学生科研展示、答辩与在线作品集

Defend one honest claim. Then turn the evidence into a page a future reviewer can open.

60-minute capstone · ≤ 6-slide talk · 2:00 Q&A · public-safe portfolio + resume link



HANDOFF · CLASS 14 → CLASS 15

Bring evidence, not a new experiment

Frozen from Class 14

- Question and bounded claim
- Data source, version and scope
- Saved split or folds + controls
- Pre-chosen metric and full results
- Notebook, limitation and next test

Today adds

- A six-slide evidence story
- A five-minute defense-ready talk
- Glow · Grow · Next test feedback
- A portfolio project card + resume-ready link

Do not rerun until the answer looks better. If something changed after Class 14, disclose exactly what and why.



TODAY • THE ROOM CONTRACT

Students speak; the teacher tests the evidence

Presenter

Make one bounded claim, show the audit trail, stop on time, and say “I don't know” when the evidence ends.

Teacher

Ask at least two questions: one about **validity**, one about **interpretation or reproducibility**.

Peers

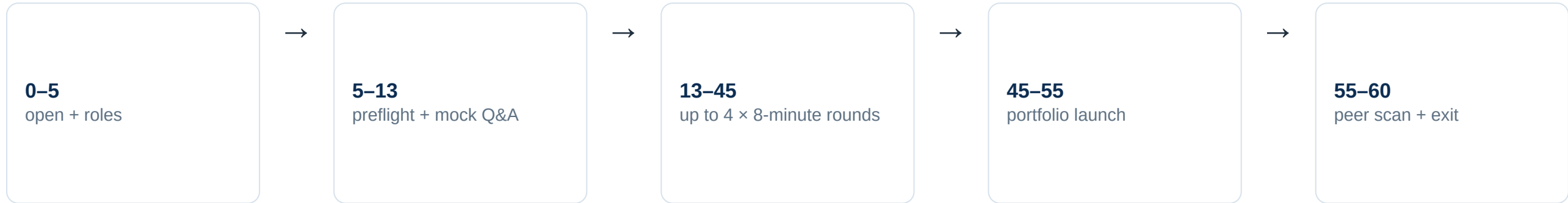
Listen for the chain of evidence and leave one Glow, one Grow and one Next test.

Privacy boundary: use a teacher-assigned presenter code. Show no full name, email, school, student ID, address, health detail, face, or raw personal record. Use aggregate or de-identified evidence only.



RUN OF SHOW · ONE-HOUR CAPSTONE

Sixty minutes, with student work protected



Fewer than four presenters

Use the recovered minutes for a longer portfolio build, rehearsal, or second-round defense question.

More than four presenters

Use parallel rooms or a second full session. Do not shrink the defense until it becomes ceremonial.



PREFLIGHT 1 OF 4 · SUPPORTED CLAIM

Quick check · tap an answer

A candidate model outperformed the baseline on the same predefined folds, but only one dataset was tested. Which headline is defensible?

- A** The candidate is best for every future dataset
- B** On this dataset and protocol, the candidate performed better; outside generalization is untested
- C** The experiment proves the model caused every improvement
- D** Remove the limitation so the conclusion sounds stronger



PREFLIGHT · THE SENTENCE TEST

Can the whole study fit in one honest sentence?

For **[population / dataset scope]**, we tested whether **[one change]** affected **[pre-chosen metric]** versus **[baseline]**, using **[saved protocol]**; we found **[bounded result]**, limited by **[largest threat]**.

Names the comparison

Change, baseline and outcome are visible.

Names the protocol

The audience knows how the test stayed fair.

Names the boundary

The claim stops where the evidence stops.



DECK RULE · MAXIMUM SIX SLIDES

Six slides, including the title

1 • Question

Title + bounded research question.

2 • Data

Source, version, n, scope and exclusions.

3 • Method

Change, baseline, controls and protocol.

4 • Results

Full comparison + uncertainty/error view.

5 • Meaning

Conclusion, limitations and rival explanation.

6 • Reproduce

Notebook/data links, AI disclosure, next test.

Six is a ceiling, not a target. Merge or cut anything that does not help the audience audit the claim.



STUDENT SLIDE 1 · QUESTION

Give the audience a question they can hold

Title formula

One change + one outcome + one scope. Prefer a question or bounded finding over “My AI Project.”

Opening line

“I tested whether ___ changes ___ for ___.” Add one sentence explaining the decision or uncertainty.

Keep off the screen

Biography, table of contents, inflated impact claims and identifying student information.

RUNNING TEACHING EXAMPLE

Does feature scaling help 5-NN classify penguin species?

A controlled comparison on complete Palmer Penguins measurements



Adelie



Chinstrap



Gentoo

One change · scaling

One outcome · accuracy

One scope · Palmer Penguins

Computed classroom example · not student work · not a claim about new datasets



STUDENT SLIDE 2 · DATA

Turn “we used data” into one visible row

Provenance

Name, source, version or collection window, license and access path.

Unit + target

Say what one row represents, what is measured and what label or outcome is predicted.

Boundary

Counts, exclusions, missingness, coverage gaps and who or what is absent.

EXAMPLE · DATA CARD

Palmer Penguins

One row = one observed penguin. The label is Adelie, Chinstrap or Gentoo.



Adelie



Chinstrap



Gentoo

342

complete penguins

4

body measurements

3

species

bill length

bill depth

flipper length

body mass

344 source rows · 2 incomplete rows excluded · Palmer Station LTER / palmerpenguins · CC0-1.0



Show two lanes so the one change is obvious

Comparison contract

Baseline · one intended change · controls · pre-chosen metric.

Protocol receipt

Saved folds, seed, model settings, preprocessing boundary and threshold rule.

Leakage defense

Fit learned preprocessing inside each training fold—never on all rows before validation.

EXAMPLE · FAIR COMPARISON

Same penguins · same 5 folds · same 5-NN · same accuracy function

Baseline

raw feature scales → 5-NN

accuracy receipt

Contender

StandardScaler → 5-NN

accuracy receipt

seed 15

stratified folds

scaler fit inside training fold

The only intended change is standardization; the evaluation rows and classifier settings stay fixed.



Quick check · tap an answer

The candidate uses a new feature set and a new train/test split; the baseline uses the old features and old split. The candidate wins. What can the presenter defend?

- A** The new features caused the gain
- B** The candidate is universally better
- C** Only that the two completed runs differ; the cause is confounded
- D** The split never matters when the metric is identical



STUDENT SLIDE 4 · RESULTS

Give one focal comparison—and keep the receipts nearby

Focal visual

Put baseline and contender adjacent. Label metric, units, sample count and evaluation protocol.

Variation

Show every fold or an honest center plus spread; keep failed and weak runs visible.

Reading sentence

Say what the chart establishes, then stop before mechanism or outside generalization.

EXAMPLE · COMPUTED RESULT

Five-fold accuracy on 342 Palmer penguins

Raw 5-NN  79.5%

Scaled 5-NN  98.5%

Observed difference +19.0 pp raw folds 76.8% · 76.8% · 82.4% · 75.0% · 86.8%

scaled folds 98.6% · 98.6% · 97.1% · 100.0% · 98.5%

Zero-based bars · same frozen folds · classroom computation, not a student submission.



Quick check · tap an answer

A candidate catches more positives but also creates more false alarms than the baseline. Which headline is honest?

- A** The candidate is simply more accurate
- B** The candidate solves the task
- C** The candidate trades precision for recall; whether that helps depends on the stated error costs
- D** Hide precision because recall was the chosen metric



Conclusion + limits belong on the same screen

Supported conclusion

Answer the question using the observed direction, size and uncertainty.

Largest limitation

Name the threat most likely to change the result — not a generic “more data.”

Rival explanation

What else could have produced the pattern, and what evidence would separate it?

EXAMPLE · THREE DIFFERENT SENTENCES

Conclusion

On these fixed penguin folds, scaled 5-NN averaged 98.5% versus 79.5%.

Boundary

One dataset, one classifier setting and one fold assignment do not establish a universal scaling benefit.

Next test

Repeat the frozen comparison across other datasets, k values and fold seeds.



Finish with a compact receipt, not a wall of code

Reproduction card

Code/version · environment · data identity · seed/folds · settings · saved outputs.

AI-use disclosure

What the tool generated, what you changed, and which outputs you independently reran and checked.

One next test

Name an analysis that could challenge the conclusion—not decoration that assumes it is already true.

EXAMPLE · REPRODUCIBILITY FOOTER

From data loader to both accuracy receipts

```
data: palmer-penguins.csv @ 8957207 (342 complete × 4)
features: bill length/depth · flipper length · body mass
cv: StratifiedKFold(5, shuffle=True, seed=15)
model: KNeighborsClassifier(k=5)
change: StandardScaler fitted inside each training fold
metric: accuracy · sklearn 1.9.0
```

AI-use example: AI helped draft the slide wording; the code was rerun and every displayed value was checked against saved output.

A file, cell, query or calculation should answer “where did this number come from?”



The same evidence, two reading speeds

Showcase slide	Paper section	What remains identical
1 · Question	Title + introduction	Scope and research question
2 · Data	Data / methods	Provenance, sample and exclusions
3 · Method	Methods / experiment	Baseline, controls and protocol
4 · Results	Results	Complete table, uncertainty and errors
5–6 · Meaning + reproduce	Discussion, limits, references	Bounded claim, caveat, artifacts and AI use



TALK MODEL · AIM 4:30

Six moves, with thirty seconds of safety margin

0:00–0:30

Question + why it matters

0:30–1:10

Data, source and scope

1:10–2:15

Design, controls and metric

2:15–3:30

Results + uncertainty/errors

3:30–4:10

Conclusion + largest limit

4:10–4:30

Reproduce + next test

Hard stop at 5:00. Rehearse to 4:30 so one pause, transition or clarification does not erase the ending.



DEFENSE · ANSWER STRUCTURE

Answer the question before adding context

1 · Listen

repeat the question if needed



2 · Answer

one direct sentence



3 · Evidence

point to result or protocol



4 · Boundary

what remains unknown

Useful starter

“From this experiment, I can say ___ because ___. I cannot yet tell ___.”

If challenged

Thank the question, separate evidence from inference, and propose the next discriminating test.



PREFLIGHT 4 OF 4 · UNKNOWN ANSWER

Quick check · tap an answer

A teacher asks whether the result holds for a subgroup the project never analyzed. What is the strongest response?

- A** Yes, because the overall score was good
- B** Probably — then change the subject
- C** I do not know from this study; I would predefine the subgroup test and evaluate it without reusing the final test set
- D** Delete that subgroup from the dataset



TEACHER CARD · MINIMUM TWO QUESTIONS

One validity question + one meaning/reproduction question

Lane A · validity

- What changed, and what stayed fixed?
- How did you prevent leakage?
- Why does the split match real use?
- What baseline had to be beaten?
- Was the metric chosen before results?

Lane B · interpretation / reproduction

- What evidence would reverse your conclusion?
- Which limitation matters most?
- Can another student rerun this exactly?
- What AI output did you verify or reject?
- What is the next test, and why?

Ask at least one from each lane. Questions test ownership of the study, not trivia or public-speaking style.



PEER LISTENER CARD

Glow · Grow · Next test

Glow

The strongest **supported** claim or clearest evidence link.

Grow

One place where question → method → result → conclusion was unclear.

Next test

One feasible experiment that could challenge the claim or address its largest limit.

"Glow: I trusted ____ because _____. Grow: clarify _____. Next test: hold ____ fixed and test _____."



ASSESSMENT · ALIGNED TO CLASS 14

Showcase + defense rubric · 100 points

Evidence category	Points	Full-credit signal
Question + bounded claim	15	Specific, measurable and no broader than the design
Data	10	Source/version/n/unit/exclusions and privacy-safe display
Experiment fairness	20	Credible baseline, one change, controls, saved split/folds and leakage defense
Results + uncertainty	20	Complete comparison, pre-chosen metric, spread/errors and no cherry-picking
Interpretation + limits	15	Evidence-based meaning, rival explanation, largest threat
Reproducibility + AI disclosure	10	Rerunnable artifacts and verified, specific AI-use record
Slides + delivery + timing	5	≤6 slides, readable, aim 4:30 and stop by 5:00
Defense / Q&A	5	Direct, evidence-linked answers with honest boundaries

Total: 100. A null result can earn full credit; unsupported certainty cannot.



PRESENTER · SIXTY-SECOND PREFLIGHT

Ready means the evidence can survive the projector

- Deck has **six slides or fewer**, title included
- Presenter code only; no student PII
- Every number comes from the final notebook
- Axes, units, denominators and labels are readable
- Limitation and next test are visible

- Notebook/data links open with correct permissions
- AI disclosure names verification actions
- Talk rehearsed to **4:30**
- One validity answer prepared
- One interpretation/reproduction answer prepared

If a screen contains private data, close it before sharing. A screenshot is not a privacy plan.

[Open printable showcase worksheet ↗](#)



LIVE CONTROL · PRESENTER

4:30 aim

0:00
question



1:00
method



2:15
results



3:30
meaning



4:30
finish

Hard stop: 5:00. At 4:30, finish the current sentence and land the limitation/next test.



LIVE CONTROL · TEACHER DEFENSE

2:00 Q&A

Question 1 · validity

Design, baseline, controls, leakage, split or metric.

Question 2 · interpretation / reproduce

Claim boundary, limitation, rerun, AI verification or next test.

Two questions are the minimum. Keep answers direct so both lanes are tested.



LIVE CONTROL · TRANSITION

1:00 feedback + switch

Peers

Finish Glow · Grow · Next test.

Teacher

Capture rubric evidence; do not write from memory later.

Next presenter

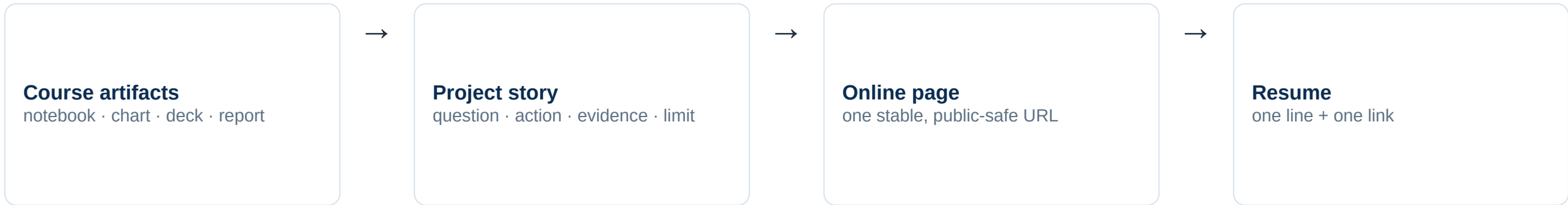
Open deck, confirm presenter code, start on title slide.

No public ranking. Return feedback privately or through the class's approved channel.



AFTER THE DEFENSE · KEEP THE WORK ALIVE

A submission ends a class.
A portfolio starts the next conversation.



The portfolio is not a trophy shelf. It is a short route from “What did you do?” to evidence another person can inspect.



AUDIENCE TEST · DESIGN BACKWARD

What should a reviewer know in ten seconds?

1 · Your direction

“I investigate how data and models answer questions in ____.”

2 · Your best project

A title plus one sentence explaining the question and what you built.

3 · Your evidence

One readable result, one artifact link and one honest limitation.

Ten-second test: cover your name. Can a stranger still tell what problem you studied, what you did, and why the result is credible?



CHOOSE A LANE · SAME EVIDENCE, DIFFERENT CONTAINER

Pick the simplest platform you can keep updated

Fastest · no code

Google Sites, Notion or Canva website.

Best when tomorrow's goal is a working link.

Code-forward

GitHub Pages.

Best when the repository itself is part of the evidence.

Private / reviewer-only

PDF or access-controlled page.

Use when age, policy, licensing or data sensitivity makes public sharing inappropriate.

Platform choice is not the achievement. A clear project card on a simple page beats an elaborate site with no evidence.



ONE-PAGE ARCHITECTURE · FOUR READING DEPTHS

Make the page useful at a glance—and on inspection

Top · 10 seconds

Professional display name or approved alias · one-line focus · one featured project.

Project card · 30 seconds

Question · your role · methods/tools · supported result · limitation.

Evidence · 2 minutes

Chart with caption · notebook/repository · short report/deck · reproducibility note.

Next step · 10 seconds

What you learned · next test · safe contact or application route.



CURATE · DO NOT UPLOAD EVERYTHING

Choose three artifacts that prove different skills

Artifact	Course evidence	What it proves
One focal visual	Class 9 or Class 14 chart/result	You can find and explain a pattern honestly
One reproducible artifact	Notebook, code or experiment record	You can make the result inspectable
One communication artifact	Deck, short paper or research map	You can explain the work to another person

Remove duplicates, broken drafts, private data and unexplained files. Curating is part of authorship.



FEATURED PROJECT · THE SIX-LINE CARD

Turn the mini-study into a reviewer-facing story

1. **Question:** What did you try to learn?
2. **Role:** What did you personally do?
3. **Method:** Data, baseline and one controlled change.

4. **Evidence:** What did the fair comparison show?
5. **Boundary:** What can it not establish?
6. **Next test:** What would reduce the uncertainty?

Write for a reader who has not taken this class. Expand acronyms once. Replace “we used AI” with the exact task, tool and verification.



EVIDENCE GALLERY · ONE IMAGE, THREE RECEIPTS

Show less—and explain each item more

Focal result

One chart or table with metric, units, comparison and a self-contained caption.

Reproduce

Notebook/repository + data identity + environment or “how to rerun” note.

Read further

Short report or ≤ 6 -slide deck for the reasoning and limitations.

Link hygiene: descriptive link text, working permissions, stable filenames, no “click here,” no credentials, and no raw personal record.



REFLECTION · MAKE GROWTH VISIBLE

The most interesting sentence may be what changed

I first assumed...

initial model of the problem



The evidence showed...

result or failure that changed it



So I revised...

claim, method or next question

Strong reflection

Names a concrete decision, evidence and revision.

Weak reflection

“I learned a lot about AI” with no trace of what changed.



BUILD SPRINT 1 · EIGHT MINUTES

Write the top of the page + your project card

Minutes 0–2

Professional display name or approved alias + one-line focus.

Minutes 2–4

Project title + question + your personal role.

Minutes 4–6

Method + supported evidence, with scope attached.

Minutes 6–8

Largest limitation + one next test.

[Open Portfolio Launchpad ↗](#)

Your entries stay in this browser unless you download the page.



BUILD SPRINT 2 · SIX MINUTES

Add evidence, links and a growth sentence

Evidence

Choose one chart and write a caption: what it shows → why it supports the claim.

Receipts

Add repository/notebook and report/deck links; test them in a private window.

Growth

“I first assumed __; evidence showed __; so I revised __.”

If a link cannot be public safely, write “available to approved reviewers” and keep the artifact in the approved private channel.



RESUME • TURN NOUNS INTO EVIDENCE

Action + artifact + method + evidence + boundary

Designed and evaluated [project / model] on [named data] using [fair-comparison method]; found [bounded result], documented [limitation], and published a reproducible project page.

Avoid

“Used Python and AI for a machine-learning project.”

Prefer

A verb, a concrete artifact, the comparison you owned and the scope of the evidence.



LINK MAP · MAKE THE EVIDENCE REACHABLE

One stable portfolio URL, placed where readers look

Resume header

Portfolio: short readable URL beside email / GitHub / LinkedIn, if those are appropriate.

Project entry

Link the project title to the exact project card—not only the site homepage.

Applications

Use the same stable link in forms, emails and mentor introductions; test on mobile.

A QR code is optional and secondary. Always print the readable URL because scanners, PDFs and applicant systems may strip images.



BEFORE PUBLISH · PUBLIC, PRIVATE OR REMOVE

Every item needs an audience decision

Public-safe by default	Reviewer-only unless approved	Remove
Aggregate charts · original explanations · licensed/project-owned images · de-identified code	Full notebook with controlled data · manuscript under review · team artifacts · age/school details	Passwords/API keys · raw personal records · private emails · precise address/schedule · unlicensed data

Minors: follow family, school and program rules. A professional alias, private link or PDF is a valid portfolio—not a lesser one.



PEER QA · FIVE-MINUTE SCAN TEST

Review the page as a stranger, not as a classmate

10 seconds

Say the problem, the student's role and the strongest evidence—without asking the author.

2 minutes

Open every link. Check access, labels, captions, mobile readability and private information.

Final note

Write one Glow, one missing receipt and one sentence to cut or clarify.

Pass condition: the reviewer can explain the claim and its boundary, then reach the evidence in two clicks or fewer.



AFTER THE DEFENSE · FINAL PACKAGE

Submit the evidence—and launch the route to it

Research package

- Final ≤ 6 -slide deck
- Short paper or structured report
- Runnable notebook/repository
- Dataset citation + access instructions
- Environment, seed and split/fold record

Portfolio launch

- One project card and focal visual
- Results agree across page, slides and code
- Limitation + AI disclosure remain visible
- Public/private/license audit complete
- Stable URL tested on mobile and in a private window

Required for class: research package + portfolio draft. **Public publishing is optional** and must follow the student's approved privacy path.



WRAP-UP · THE RESEARCH LOOP CLOSES

The course ends. Your evidence now has somewhere to go.

Defend

One claim, small enough for the design to support.

Publish safely

One page that connects the claim to inspectable evidence.

Carry forward

One resume line, one stable link and one next test.

Exit line: “I studied ____; the evidence supports ____; you can inspect it at ____; next I would test ____.”

Public, private or reviewer-only—the audience choice is part of responsible research.